DEPI SOFTWARE TESTING TRACK — FINAL PROJECT

Key Performance Indicators (KPIs)

Project Performance Targets, Quality Metrics, Measurement Formulas, Data Sources, and Review Governance

Project	Banking Transaction System	**Document**	Key Performance Indicators (KPIs)
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Part 1. Introduction & Purpose

Establishing objective, measurable engineering and quality benchmarks for the Banking Transaction System.

Key Performance Indicators (KPIs) are quantifiable criteria used to evaluate the Banking Transaction System's schedule progress, software quality, test completeness, and readiness for final delivery. Defining KPIs during the initial planning phase establishes unambiguous performance targets before technical implementation and testing begin. This document specifies the five core project KPIs, their mathematical measurement formulas, target thresholds, data sources, and review cadence.

Part 2. Project KPIs Catalogue

The five core performance and quality indicators governing project execution and release sign-off.

ID	KPI Name	Purpose	Measurement Formula / Method	Target Benchmark
KPI-01	Schedule Adherence	Evaluate whether planned weekly engineering and QA tasks are completed according to the agreed 8-week schedule.	(Number of tasks completed on time / Total tasks due in the period) × 100	**At least 90%**
KPI-02	Requirements Test Coverage	Ensure that every testable functional and non-functional requirement is verified by at least one documented test case.	(Number of testable requirements covered by test cases / Total testable requirements) × 100	**100%**
KPI-03	Test Execution Completion	Measure the completion of all planned manual and automated testing activities prior to milestone sign-off and final release.	(Number of planned test cases executed / Total planned test cases) × 100	**100% of planned test cases executed (or formally accounted for) before release**

ID	KPI Name	Purpose	Measurement Formula / Method	Target Benchmark
KPI-04	Critical & High-Severity Open Defects	Monitor unresolved high-impact security, money-integrity, or functional defects prior to final system delivery.	Count of open Critical and High-severity defects recorded in the Jira defect tracker.	**Zero (0) open Critical or High-severity defects at release**
KPI-05	API Response Time (Latency)	Verify that backend REST API operations satisfy the system's non-functional performance requirement under normal load.	Measure REST API endpoint response times using Postman Runner with seed data and evaluate the 95th percentile (P95).	**P95 response time below 500 ms**

Part 3. Measurement Methods & Data Sources

Authoritative tools and repositories used to collect and verify each KPI metric.

KPI ID & Name	Authoritative Data Source	Measurement Frequency
KPI-01: Schedule Adherence	Project Plan (8-Week WBS) and Jira Sprint Board task-tracking records	Weekly (during the team sync meeting)
KPI-02: Requirements Test Coverage	Requirement Gathering specification and Requirement Traceability Matrix (RTM)	During test planning and whenever requirements or test cases are updated
KPI-03: Test Execution Completion	Manual test execution records, JUnit/Surefire reports, Postman results, and Selenium reports	During each testing cycle and prior to final release
KPI-04: Critical & High-Severity Open Defects	Jira Defect Tracker and bug triage logs	Continuously during testing (Weeks 5–7) and before final release
KPI-05: API Response Time	Postman Collection Runner automated performance test results	During API integration and performance testing (Weeks 5–7)

Part 4. KPI Monitoring, Review & Corrective Actions

Governance checkpoints and remediation protocols when a metric deviates from its target.

KPI Name	Milestone Review Checkpoint	Corrective Action Protocol if Target Is Missed
Schedule Adherence	Monitored weekly throughout development (Weeks 1–8)	Reallocate blocked tasks across team members or activate the agreed feature descoping order (cutting Could/Should items first) to protect core Must deliverables.

KPI Name	Milestone Review Checkpoint	Corrective Action Protocol if Target Is Missed
Requirements Test Coverage	Reviewed as requirements and test cases are documented (Weeks 2–5)	Block test plan sign-off until missing test cases are authored and linked in the Requirement Traceability Matrix (RTM).
Test Execution Completion	Evaluated during active testing phases (Weeks 5–7)	Prioritize execution of Must-have and high-risk financial/security test suites; quarantine flaky UI tests and verify via API/manual tests.
Critical & High-Severity Open Defects	Evaluated continuously during testing and at Milestone 4 (Week 7)	Freeze new feature development immediately; assign senior developers to reproduce, fix, and regression-test all Critical/High defects before demo rehearsal.
API Response Time	Evaluated during API performance testing (Weeks 5–7)	Profile slow SQLite queries, add missing database indexes on foreign keys/account numbers, and optimize N+1 JPA query patterns.

Part 5. Operational Governance & Baseline Status

How KPI targets transition from planning baselines to active release gates.

These five KPI targets serve as the formal quality gates for the Banking Transaction System. Actual KPI measurements are recorded by the QA Lead in the weekly status log and Jira dashboard as development and testing activities progress. No release or final demonstration sign-off is complete until KPI-02 (100% coverage), KPI-03 (100% execution), KPI-04 (0 Critical/High bugs), and KPI-05 (P95 < 500 ms) are verified with concrete testing evidence.